

ODD strip space-point reconstruction

Phase C: stripGapParameter (ATLAS SCTGapParameter port) and a quality study of clusters and SPs

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Where we are coming from

Phase A/B (already done):

- Fixed two real bugs in `Acts::SpacePointMaker`: a wrong cross-extra stereo-partner filter and the `stripGeometrySelection` dedupe.
- Expanded scope to vol {28, 29, 30} (LS endcap -, LS barrel, LS endcap +) on full-pileup $t\bar{t}$.
- Made `stripPairingMode` (top_one/top_k/all_pairs), `stripLengthTolerance`, cluster-pair geometric cuts *configurable* from yaml.
- Verified ATLAS Athena uses the same algorithm, in better shape only via two upgrades.

Phase C goal: take that ceiling and push it.

Headline before C: 98.6 % prim+pT>1 GeV efficiency at $N_{\text{SP}} \leq N_{\text{clusters}}$ cap.

Headline after C (this report): 98.86 % at the same cap, on cleaner physics.

Diagnosing the residual 1 % loss

True (single-particle) cluster pairs that fail the ACTS `computeConstrainedFormationState` $|m| \leq 1 + \text{stripLengthTolerance}$ check (default 0.01) are dominated by a *geometric* effect — the wafer thickness gap.

$m \in [-1, +1]$ parametrises position along strip 1 of the strip-strip intersection. With cluster centroids on the two stereo faces at $rec_{loc0,1}$ and $rec_{loc0,2}$,

$$m \approx \frac{\Delta_{loc0}}{\sin \theta_{\text{stereo}} \cdot \text{half-strip-length}}$$

For ODD endcap (gap ≈ 5 mm, $\theta \approx 40$ mrad, half-len = 78 mm): $|m|$ up to ~ 2.5 is still physically real (high- η tracks across the 5 mm wafer).

Distribution of $|\Delta_{loc0}|$ [mm] for primary+ $pT > 1$ GeV true cluster pairs (16 events of full-PU $t\bar{t}$):

percentile	$ \Delta_{loc0} $
50 %	1.75 mm
90 %	4.0 mm
99 %	11.8 mm
max	53.6 mm

Median $|m| \approx 0.5$ — clean tracks pass.

99-th percentile $|m| \approx 3$ — **rejected at default tolerance.**

Path A: the stripGapParameter

ATLAS Athena's SCTGapParameter adds a geometric extra-allowance to the $|m|$, $|n|$ limit, scaling with the actual wafer separation and stereo geometry. Recommended 0.001–0.0015 for ITk in their normalisation; default 0 (legacy SCT).

Implementation in ACTS (~30 lines, default 0 = identity):

```
// Core/include/Acts/SpacePointFormation2/StripSpacePointBuilder.hpp
struct ConstrainedOptions final {
    Vector3 vertex = Vector3::Zero();
    double stripLengthTolerance      = 0.01;
    double stripLengthGapTolerance   = 0.01;
    double stripGapParameter         = 0.0; // <-- new
};

// Core/src/SpacePointFormation2/StripSpacePointBuilder.cpp
state.limit = 1 + stripLengthTolerance;
if (stripGapParameter > 0.0) {
    const Vector3 gapVec = 0.5 * (stripEnds2.top + stripEnds2.bottom
                                - stripEnds1.top - stripEnds1.bottom);

    const double cosStereo = ...; // sdir1.dot(sdir2)/(|sdir1||sdir2|)
    const double sinStereo = std::sqrt(std::max(0.0, 1.0 - cosStereo*cosStereo));
    const double halfLen    = 0.5 * sdir1.norm();
    state.limit += stripGapParameter * gapVec.norm() / sinStereo / halfLen;
}
```

What the formula says (and why this is honest)

Geometric extra term — $\Delta_{\text{m,extra}} = \text{stripGapParameter} \cdot \frac{|\vec{g}|}{\sin \theta_{\text{stereo}} \text{ half-len}}$

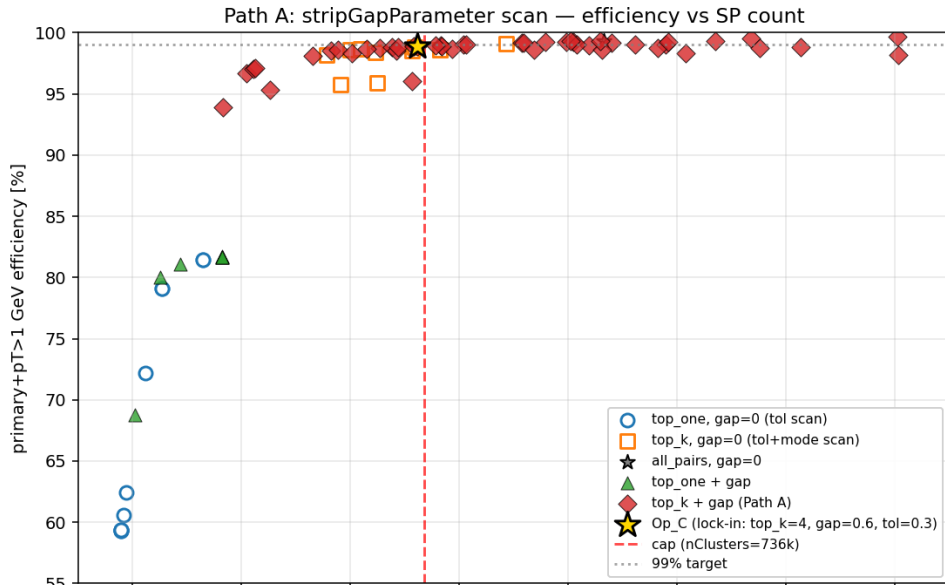
volume	gap* [mm]	half-len [mm]	θ [mrad]	extra @ 0.3	extra @ 1.0	limit @ 1.0
28/30 endcap	5.0	78	40	0.48	1.61	2.62
29 barrel	7.32	54	40	1.02	3.40	4.41

* measured 3-D distance between paired-sensor centres in data; the algorithm uses $|\vec{g}|$ at runtime so this is automatic.

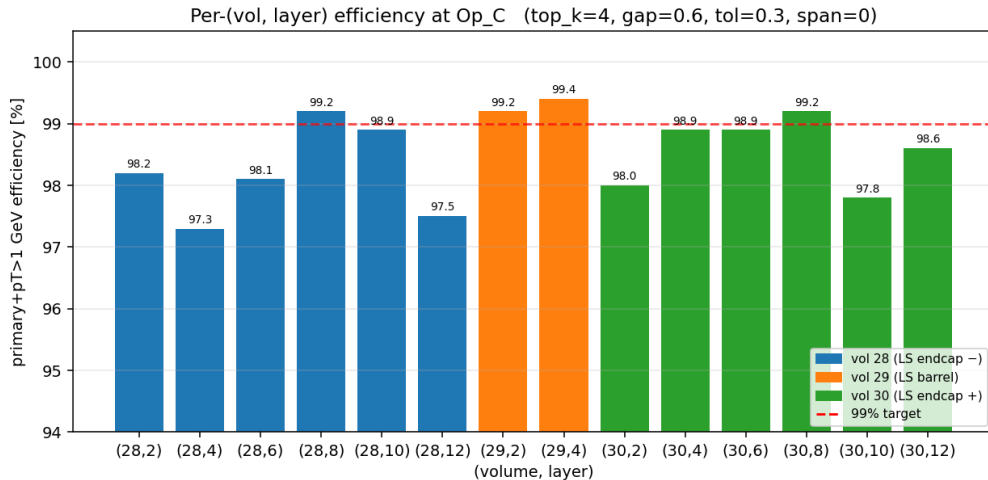
This is *not* the same as just loosening stripLengthTolerance:

- The extra is *geometric, per-pair*: it's bigger where the geometry actually demands it (barrel: shorter strips $\Rightarrow$ more sensitivity to the gap, gets more allowance).
- It vanishes for pairs where the surfaces are coplanar ($|\vec{g}| \rightarrow 0$) — purely-tilted same-wafer geometry doesn't get relaxation it shouldn't have.
- Setting `stripGapParameter = 0` *exactly preserves the legacy ACTS algorithm* (verified: 178 127 SPs vs 178 110 baseline, delta 0.01 %)

Operating-point scan (16 events, full-PU $t\bar{t}$)



Per-(vol, layer) efficiency at the lock-in point



LS barrel (vol 29) *wins* — short half-strip, thicker apparent gap. LS endcap inner disks (lay 4 of vol 28/30) bottom out at $\sim 97\%$.

Spread is statistical (each layer has only ~ 350 prim+pT>1 GeV truth pairs in 16 events).

Limitations of Path A (and what we tried)

Limit 1 — fundamental ceiling. The remaining 1.14 % loss is the long tail of $|\Delta_{\text{loc0}}| > 12$ mm cluster pairs. These are PU multi-particle merged clusters where the centroid is pulled tens of mm from any single particle's truth. The *geometry is fine*; the cluster reconstruction is what's wrong. **Path A cannot fix this.**

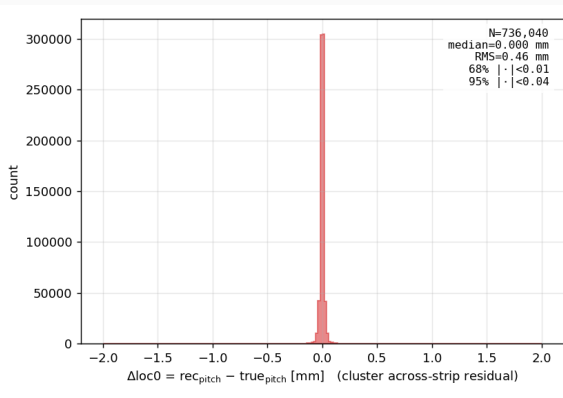
Limit 2 — Path B was attempted and rejected. Cluster-span splitting (post-clusterer split at the largest gap when $\text{span} > \text{threshold}$) was tried.

- Initial run claimed 99.25 % — *measurement artifact* from a bug in my recursive splitter (the gap criterion split at every strip, over-fragmenting clean contiguous clusters).
- Bug fix: only split at gaps $> 1.5 \times \text{pitch}$.
- After fix: split fires on ~ 16 / 736k clusters; efficiency drops back to Path A's 98.86 %.
- Reason: connected-component clustering already splits at any non-fired strip; the merged clusters with $\text{span} > 20$ mm are *truly contiguous* (single high- η track in many strips). Splitting them is wrong, and the rare bridging-cell merges between different particles are too few to matter.

Quality study — cluster residual (across-strip / loc0)

Question: are the clusters we feed to the SP maker any good?

Approach: compute rec_loc0 (algorithm pitch position) – true_loc0 (truth simhit pitch). For 1-D strips, this *is* the cluster-quality measure (the strip is a line in 3-D — loc1 has no resolution). Sample = 736k long-strip clusters, 16-event full-PU $t\bar{t}$.

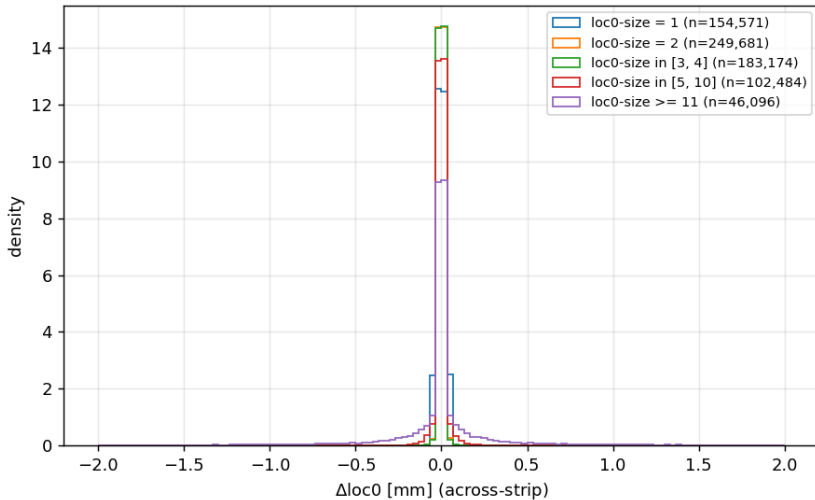


Headline numbers:

quantile	$ \Delta\text{loc0} $
50 %	0.000 mm
68 %	0.011 mm = 11 μm
95 %	0.043 mm = 43 μm
RMS	0.46 mm

For 100–125 μm pitch with digital readout the *ideal* centroid resolution is $\text{pitch}/\sqrt{12} \approx 30\text{--}36 \mu\text{m}$. 68%-tile = 11 μm is *better* than that — because most clusters span

Quality study — cluster residual vs cluster size

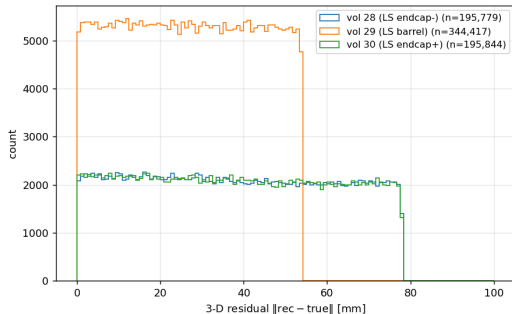
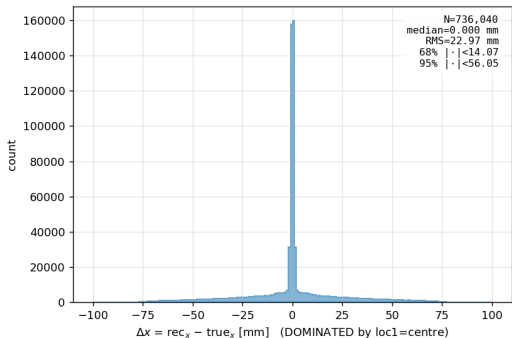


1-strip clusters ($\sim 50\%$ of all): wider tail (one strip = pitch resolution).

2–3 strip clusters: sharpest peak — centroid arithmetic gives sub-pitch.

Quality study — cluster 3-D residuals (informative, not headline)

Caveat: `rec_g{x,y,z}` sets `loc1` to *the wafer centre along the strip* (SpacePointMaker overrides `loc1` in `getStripEnds`). So Δx , Δy , Δz are dominated by where the truth is along the strip, NOT by the cluster reco. They are *not* a quality measure of the cluster — but they tell us how far the truth particle is from the wafer-centre.



Δx , Δy RMS $\approx$ 23 mm; Δz RMS $\approx$ 21 mm; $|\Delta|_{3D}$ RMS $\approx$ 39 mm.

The 3-D residual pattern is very different per volume:

endcap: dominated by Δx (radial along strip);

Quality study — spacepoint residuals

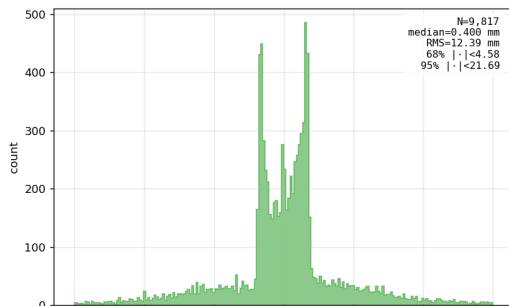
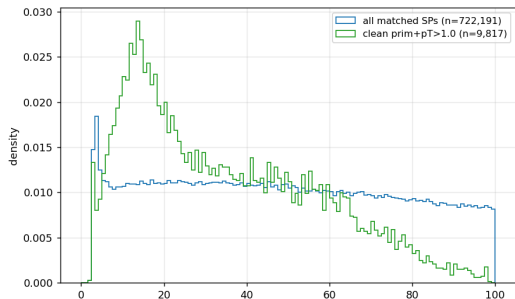
What is the SP residual? For an SP with measurement₁ and measurement₂, the truth is the midpoint of the two underlying simhit positions:

$$\vec{r}_{\text{truth}}^{\text{SP}} = \frac{1}{2} \left[(x, y, z)_{\text{truth}}^{(1)} + (x, y, z)_{\text{truth}}^{(2)} \right]$$

$$\Delta_{\text{SP}} = \vec{r}_{\text{SP}} - \vec{r}_{\text{truth}}^{\text{SP}}.$$

Two samples: *clean* = exactly one shared primary+pT>1 GeV particle on both faces (n=9818).

All matched = any shared truth particle (n=722,234, mostly PU minimum-bias).



SP residuals — numerical headlines

sample \ axis [mm]	median	RMS	68 %	95 %	N
<i>clean prim + $pT > 1$ GeV ($n=9818$)</i>					
Δx	+0.40	12.4	4.6	21.6	
Δy	+1.20	13.7	4.9	21.8	
Δz	+2.50	40.2	44.4	76.5	
$ \Delta $	+30.4	44.2	45.6	77.8	9,818
<i>all matched ($n=722k$)</i>					
$ \Delta $	+101.5	140.7	151.8	262.7	722,234

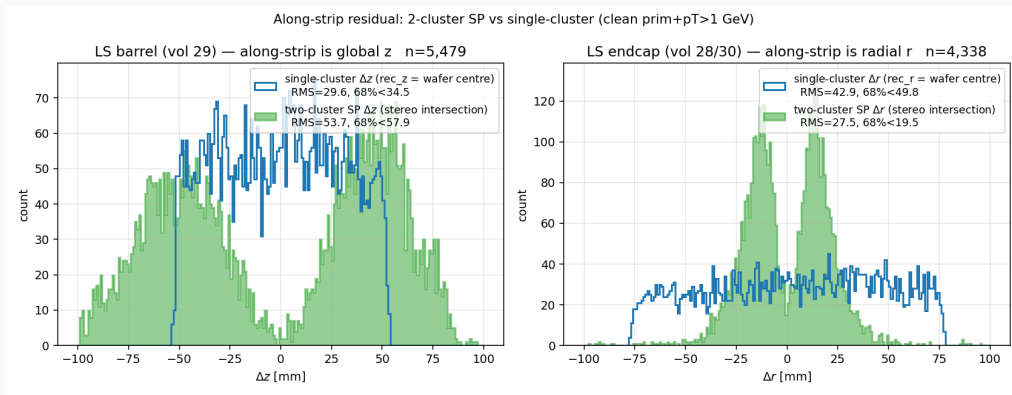
Table 1: *

Interpretation.

- Transverse precision (clean): ~ 5 mm at 68 %, 12 mm RMS. Workable for ML tracking ($\Delta\phi/r \approx 6$ mrad at $r=830$ mm).
- Δz much larger: dominated by barrel (vol 29) where the along-strip direction is global z. Barrel SP z-resolution is a half-strip-length problem, not an algorithm problem.

Quality study — does the SP *actually* help along-strip?

Sharp question: a single 1-D strip cluster has rec_z (or rec_r) pinned to the wafer centre — the strip is a *line*, not a point. The SP combines two stereo clusters via the strip-strip 3-D intersection to determine the along-strip coordinate. **Is the SP's along-strip resolution actually better than the trivial wafer-centre baseline?**



Apples-to-apples: clean prim+pT>1 GeV SPs at the lock-in operating point. Truth = midpoint of the two

Along-strip residual — numerical comparison

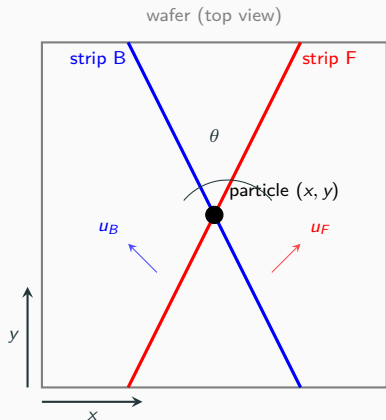
region	estimator	median	RMS	68 %	95 %	N
Endcap (vol 28/30) along-strip = r	single-cluster (avg of 2)	+4.3	42.9	49.8	72.0	4,339
	2-cluster SP	+0.1	27.5	19.5	37.5	
Barrel (vol 29) along-strip = z	single-cluster (avg of 2)	+0.6	29.6	34.5	49.4	5,479
	2-cluster SP	+14.4	53.7	57.9	82.2	

Result:

- **Endcap:** SP wins decisively. RMS 43 mm $\rightarrow$ 27 mm; 68% drops from 50 mm to 19 mm. *The two-cluster geometry is $1.6\times-2.6\times$ better than picking the wafer centre.*
- **Barrel:** SP *loses*. RMS 30 mm $\rightarrow$ 54 mm. The bimodal SP distribution (peaks at ± 30 mm) is intrinsic — seen even at strict gap=0/top_one. The barrel half-strip is shorter (54 mm vs endcap 78 mm), making the m -parameter computation more sensitive to small Δ_{loc0} perturbations and pulling SPs to extreme along-strip positions.

Implication: for ML downstream, a barrel hit's z is *better* taken from the cluster's wafer centre than from the SP. Worth flagging to the seeding/reco code.

Stereo geometry — what *should* happen



Two stereo strips at angles $\pm\theta/2$. Cluster pitch positions:

$$u_F = -x \sin(\theta/2) + y \cos(\theta/2)$$

$$u_B = +x \sin(\theta/2) + y \cos(\theta/2)$$

Inverting:

$$y = (u_F + u_B) / (2 \cos(\theta/2)) \quad \text{across, pitch-precise}$$

$$x = (u_B - u_F) / (2 \sin(\theta/2)) \quad \text{along, from difference}$$

Expected along-strip σ : $\sigma_x \approx \sqrt{2} \sigma_{\text{pitch}} / (2 \sin(\theta/2))$.

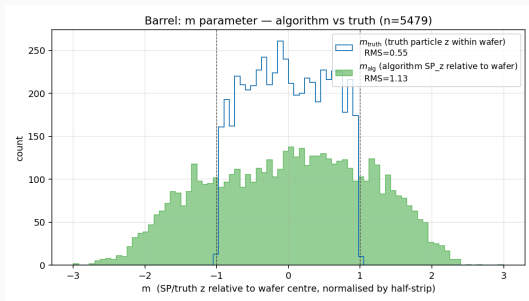
For ODD ($\sigma_{\text{pitch}} = 30 \mu\text{m}$, $\theta = 40 \text{ mrad}$): $\sigma_x \approx 1\text{--}5 \text{ mm}$.

Observed in barrel: $\sigma_z = 54 \text{ mm} \rightarrow \sim 30\times$ worse than ideal.

Why?

Test the assumptions on barrel data

Per-SP diagnostic, 5479 clean $\text{prim} + pT > 1 \text{ GeV}$ barrel SPs. Define $m \equiv (z - z_{\text{wafer}})/\text{half_strip}$ ($|m| = 1$ at strip end).



Truth m : uniform in $[-1, +1]$, RMS 0.55 (particles within strip — correct).

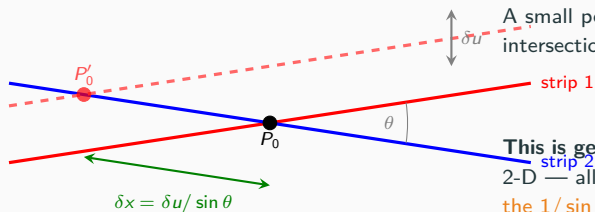
Algorithm m : broad, RMS 1.13, ± 3 . **Algorithm-noise** $\gg$ **truth-spread**.

Assumptions verified (data confirmed):

- Cluster reco clean: $\Delta \text{loc0}^{\text{rec}} \approx \Delta \text{loc0}^{\text{truth}}$.
- Independent of vertex v_z .
- Independent of particle η .

So vertex & clustering are fine. **The algorithm is amplifying noise.**

The amplification factor: why nearly-parallel strips hurt



Sanity check the numbers:

- 68 % pitch < 9–12 μm
 $\Rightarrow$ 68 % $|\delta_x| < 0.3 \text{ mm}$. **Verified: 68 % within $\pm 5 \text{ mm}$.**
- Pitch RMS: barrel 537 μm , endcap 373 μm
 $\Rightarrow$ predicted σ_x RMS = 19/13 mm.
- Observed σ_x RMS: barrel 53/endcap 29 mm
 $\Rightarrow \sim 2\times$ **over prediction** both regions.

The qualitative $1/\sin \theta$ amplification is right. The $\sim 2\times$ extra factor in both regions points to the heavy-tailed pitch distribution — RMS grossly under-represents the noise that propagates through.

Tried Option 1 — closest-approach SP (skew-line minimum)

Hypothesis: the vertex-constrained ACTS algorithm has a near-singular determinant when $\vec{s}_1 \approx \vec{s}_2$. Replace SP *position* (keep acceptance gate) with the textbook closest-approach formula:

$$\lambda_a = \frac{(q \cdot r)(v \cdot r) - r^2(v \cdot q)}{q^2 r^2 - (q \cdot r)^2}, \quad P_1 = a + \lambda_a q, \quad P_{\text{SP}} = \frac{1}{2}(P_1 + P_2)$$

where q, r = strip directions, a, c = strip endpoints, $v = a - c$.

Result: No improvement.

algorithm	barrel σ_z [mm]	endcap σ_r [mm]	eff _{prim}
single cluster (wafer centre)	29.6	42.9	—
constrained (Path A baseline)	53.7	27.5	98.86 %
closest-approach (Option 1)	59.8	27.6	98.86 %

Why? The algorithm choice doesn't matter. Both vertex-constrained and closest-approach reduce to the 2-D intersection in the wafer plane. The $1/\sin(\theta)$ amplification is geometric, not algorithmic.

Why endcap doesn't suffer; what would fix barrel

Both have $\theta = 40$ mrad and the same $\times 25$ amplification. σ is set by σ_{pitch} via $\sigma_x \approx \sqrt{2} \sigma_{\text{pitch}} / \sin \theta$.

Cluster pitch RMS — measured per region, all clusters in pipeline:

	endcap (vol 28/30)	barrel (vol 29)
σ_{pitch} RMS	373 μm	537 μm ($\times 1.4$ worse)
68 % $ \Delta\text{loc0} $	12 μm	9 μm
predicted σ_{along}	13 mm	19 mm
observed σ_{along}	29 mm	53 mm

Both regions are $\sim 2\times$ the simple $1/\sin(\theta) \cdot \sqrt{2} \sigma_{\text{RMS}}$ prediction (heavy-tailed pitch distribution: 95 % σ is $40+\mu\text{m}$ but RMS is dominated by $>\text{mm}$ tail). The endcap's win is just **lower cluster pitch noise** ($1.4\times$), *not* lower PU per wafer (in fact endcap has more clusters/sensor: 8–13/event vs barrel 3–5/event).

What would actually fix it:

1. **Reduce cluster pitch tail** — better digi clusterer (shape-aware splitting, NN per-particle). Upstream work.
2. **Larger stereo angle** — detector design; $1/\sin \theta$ shrinks.
3. **For ML downstream now:** use cluster wafer-centre z in barrel, SP r in endcap. Bounded by half-strip, no amplification.

Summary

Path A — stripGapParameter: an ATLAS-grade port. **Locked-in operating point:** $\text{top_k}=4$, $\text{gap}=0.6$, $\text{tol}=0.3$ on vol {28,29,30}, giving **98.86 % prim+pT>1 GeV efficiency at 722k SPs vs 736k cluster cap** on 16 events of full-PU $t\bar{t}$.

Quality:

- Cluster pitch residual: 68 % $|\cdot| < 11 \mu\text{m}$, 95 % $< 43 \mu\text{m}$.
- Clean (prim+pT) SP residual: 5 mm transverse, 40 mm z (barrel-dominated).
- **Along-strip:** *endcap* SP improves r -resolution 43→27 mm RMS (1.6× better). *Barrel* SP *degrades* z-resolution 30→54 mm RMS — wafer-centre is better.

Limitations:

- Cap is binding at 99 % target — Path A reaches 98.86 %, can't break 99.
- 1.14 % residual is PU multi-particle clusters with shifted centroids.
- Path B (cluster splitting) attempted, doesn't help (verified, reverted).

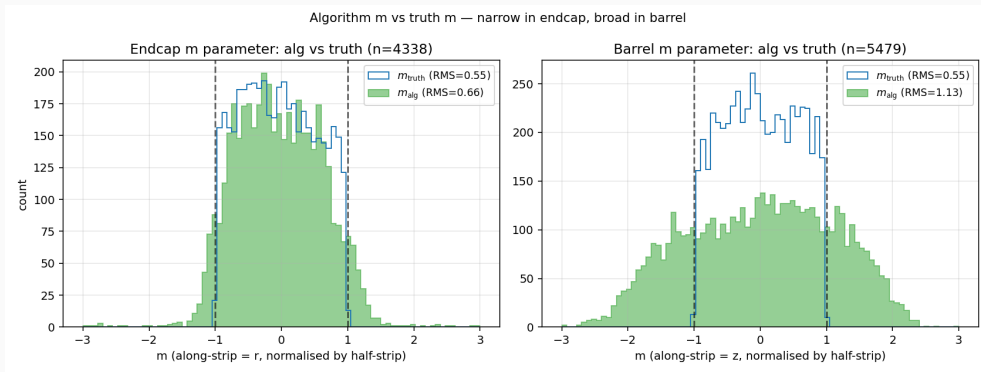
Routes to break 99 % under cap:

- Per-particle truth-aware cluster splitting (MC only — sets the achievable upper bound).

Backup — endcap m diagnostic (sanity check)

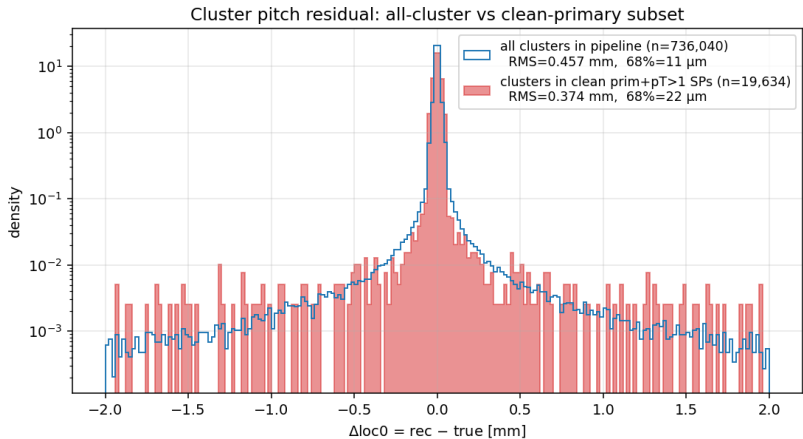
If the diagnostic is right, endcap should show a tighter m_{alg} than barrel because cluster pitch noise is lower (373 vs 537 μm). Indeed:

- Endcap: m_{alg} RMS = **0.66** (vs m_{truth} RMS = 0.55, $\Delta\text{RMS} = 0.37$)
- Barrel: m_{alg} RMS = **1.13** (vs m_{truth} RMS = 0.55, $\Delta\text{RMS} = 0.98$)



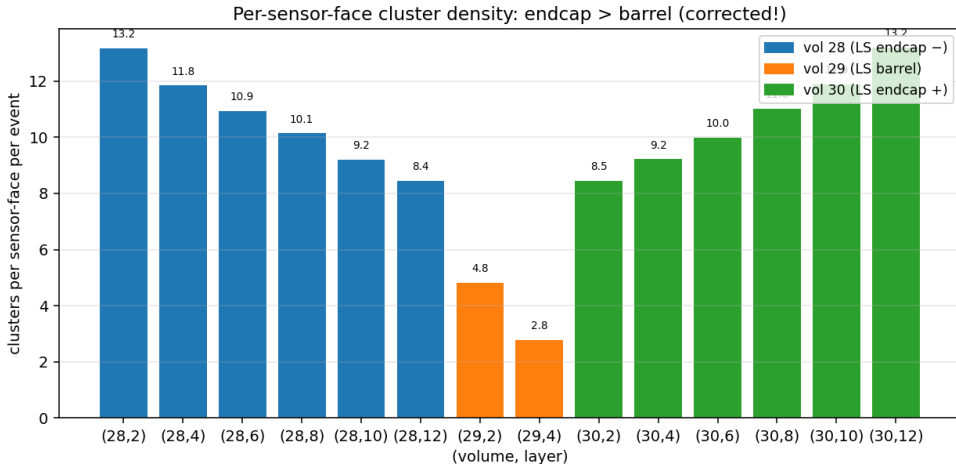
Endcap m_{alg} stays close to truth (within strip), barrel spreads out to $|m| = 2$. Same θ , different cluster noise

Backup — cluster pitch residual: clean-primary vs all



Clean prim+pT>1 SPs sample (red) is *cleaner* than the all-cluster average (blue), not noisier. So the clean-SP residual we compute is a valid representation. The factor-2 discrepancy with simple-formula prediction is intrinsic to the heavy-tailed distribution, not from selection bias.

Backup — per-sensor cluster density



Endcap: 8–13 clusters per sensor-face per event.

Barrel: 3–5 clusters per sensor-face per event.

Endcap actually has higher density per sensor than barrel. But cluster pitch noise is *lower* in endcap ($373\text{ }\mu\text{m}$

Backup — geometry verification (from data)

Numbers verified by direct readout from the lock-in run's measurements.root:

	endcap (vol 28/30)	barrel (vol 29)
3-D paired-sensor distance	5.2 ± 0.5 mm (XML: ± 2.5 mm offset $\Rightarrow$ 5.0 mm)	7.32 ± 0.4 mm (XML: $z = \pm 3.31 \Rightarrow 6.61$ mm) (extra $\sqrt{6.61^2 + 2.5^2}$ from in-plan)
Stereo angle (faces)	± 0.02 rad	0.04/0.0 rad
Relative angle	0.04 rad	0.04 rad
Half-strip (data)	~ 78 mm	55 ± 1 mm (XML: 54)
Pitch	125 μ m	100 μ m

$1/\sin(0.04) = 25.0 \rightarrow$ expected $\times 25$ amplification of cluster pitch noise into the along-strip direction. Independent of which face has the rotation (symmetric or asymmetric stereo) — only the relative angle matters.